Design, Build, and Test a PCMA Joint

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Abstract

Portable Coordinate Measuring Arms (PCMAs) are a handheld, metrology instrument used to provide adaptable 3D measurement capabilities for in-line and on-site metrology. However, current joints lack position-holding abilities, due to manual operation. This introduces dynamic positional instability and dynamic structural deformation during measurement.

This project derives technical performance measures (TPMs) from a generalised error-propagation model to design, build, and test an electromagnetic brake integrated with a PCMA joint that maintains measurement accuracy.

Introduction

Background

- PCMAs operate via multi-axis arm segments equipped with high-resolution angular encoders to capture 3D coordinates.
- Manual operation of the PCMAs causes positional instability and structural deformations because the joints cannot fix their positions independently.

Aim

Design, build and test and PCMA joint with position-holding capabilities without jeopardising its accuracy and ease of movement.

Objectives

Generalise the Denavit-Hartenberg (D-H) model, design and test key joint components, and integrate test results into the model to assess feasibility.

Methods and System Design

- **Kinematic Modelling:** Generated a six-DOF D-H model based on the Faro QuantumX arm to map spatial coordinate transformations (see Figure 1).
- **Error Analysis:** Modified and generalised an error-propagation model, first modelled by Stellenbosch Alumnus, Leon Hall, in MATLAB to create uncertainty distributions across a sampled workspace (ranging from 56 to 74 microns).

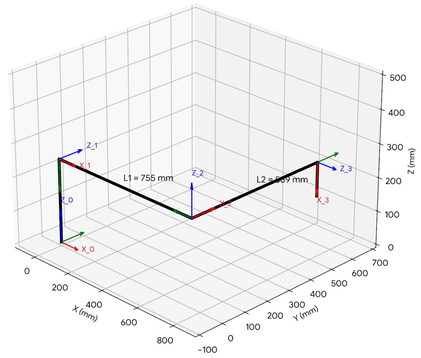


Figure 1. D-H model of Faro QuantumX arm.

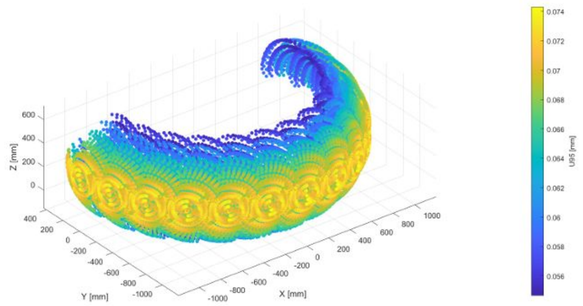


Figure 2. Error-Propagation Model

- **Concept Selection:** Generated the concepts shown in Figure 3 below. Through evaluation, the PCMA joint with an electromagnetic brake (Figure 3c) was chosen for its superior load capacity and positional stability.

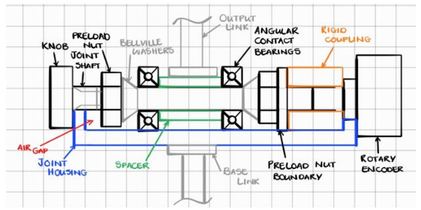


Figure 3a. PCMA Joint with Locking Dial

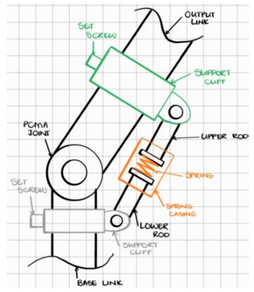


Figure 3b. PCMA Joint with Counterbalance Spring

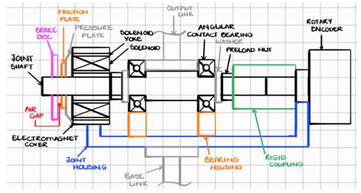


Figure 3c. PCMA Joint with Electromagnetic Brake

Figure 3. Generated Concepts

Design Refinement: The electromagnetic brake design was refined into an assembly utilising an optical encoder (GTS06-OC-RA1000B-2M), RS PRO 2-row ball bearings, an IRF640N MOSFET low-side switch controlled by an STM32 Nucleo microcontroller, and an XL7015 step-down buck converter.



Figure 4a. PCMA Joint with Electromagnetic Brake Assembly

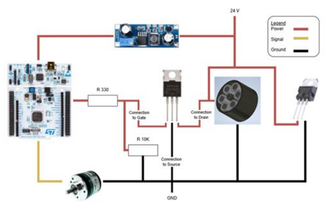


Figure 4b. Wiring Diagram of electronics implementation

Figure 4. Refined PCMA design

Discussion and Prototype Evaluation

- **Prototype Implementation Scope:** Due to project timelines, material availability and workshop capability, manufacturing and testing focused on constructing key components of the joint design, namely, the electromagnet.
- **Fabrication Constraints:** Working within workshop capabilities required balancing theoretically requirements against practical manufacturing tolerances, successfully validating core mechanical and electrical operation while highlighting areas for future scale-up.

Future Directions

- **Ongoing Performance Verification:** Conclude full experimental validation of the prototype's holding force capacity under operational cycles as seen in the setup depicted by Figure 5.

Figure 5. Testing Setup

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